

Top Ten Open Source Tools for Mathematicians

It's that time of the year when lists of all kinds are being made. So why not a list of open source tools to be used by mathematicians? The authors of this article have sifted through a number of tools and have come up with a selection that will be of immense use to mathematicians.



Over the past many years, we have gone through a large number of ‘Top Ten’ lists on various topics, both technical and non-technical. Some of them are quite interesting and others boring to the core. For example, ‘The top ten Linux distributions of all time’ is definitely an interesting read, whereas ‘The top ten things to do with a glass of water’ is downright useless, if not boring. Yes, the Internet is full of interesting, boring and stupid Top Ten lists, but we never came across a list enumerating open source tools used by mathematicians—professionals as well as academicians. So, we are listing out the top ten open source tools that are much used by mathematicians.

bc

The utility called bc (basic calculator) is an arbitrary-precision calculator language. This utility is often called the bench calculator and can be invoked with the command `bc` on a terminal. The two important versions of bc currently used are POSIX bc and GNU bc. GNU bc is available as free and open source software licensed under the GNU General Public License. The decision to include a tool like bc whose development started in 1975 in such a list might be a bit controversial. Of course, there are far more powerful tools available to choose from. Yet, we chose it because, first, it is an arbitrary-precision calculator limited only by the available memory of the host system and not limited to 8, 16, 32 or 64 bits of precision. This, really, is an important

thing and if you have any doubts, ask those programmers working in the field of scientific computing, who are plagued by the proper representation of floating-point numbers and how difficult it is to tame them.

Another reason is that though it looks very simple, bc is a Turing complete language, making it as powerful as programming languages like C and Java—at least in theory. The final reason for choosing bc for this list is that you don’t need a gun to kill a fly. On many occasions, it is a case of overkill to use powerful tools like Scilab or SageMath for simple numeric computations. Figure 1 shows the output of a division operation using bc with various levels of precision.

Scilab

Scilab is an open source, cross-platform numerical computational tool and a numerical programming language licensed under the GNU General Public License. It can be used for signal processing, statistical analysis, image processing, fluid dynamics simulations, numerical optimisation, etc, often with the help of toolboxes like Image Processing Toolbox, Wavelet Toolbox, etc. Scilab is an open source alternative to MATLAB, another very popular but proprietary numerical computing environment. The two tools resemble each other to some extent, so that often a person skilled in one will find it easy to migrate to the other. A comparison of Scilab and MATLAB is clearly out of the scope of this article, but there is at least one area where